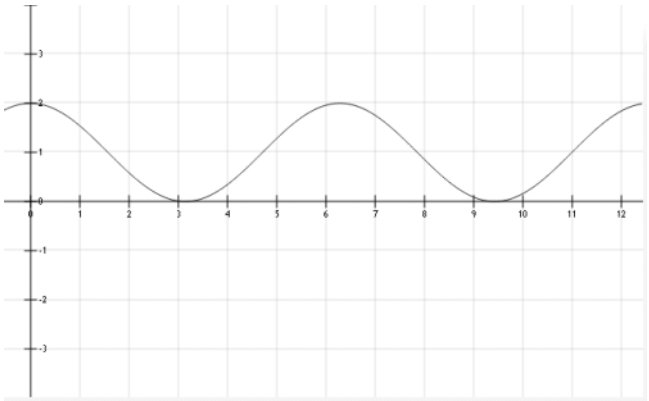
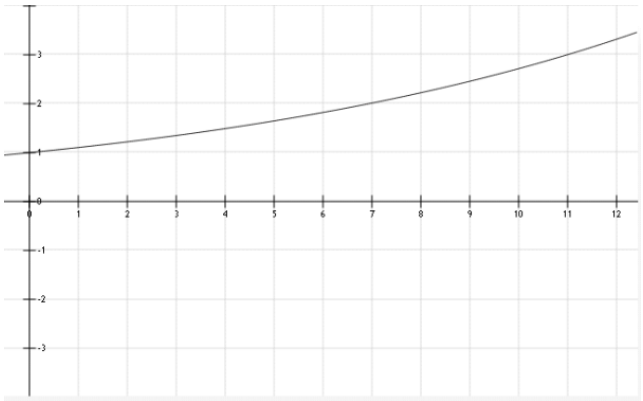
Sec 6.5 (Average Value of A Function)

Average of a finite set of numbers: $\{y_0, y_1, y_2, \dots\dots, y_n\}$

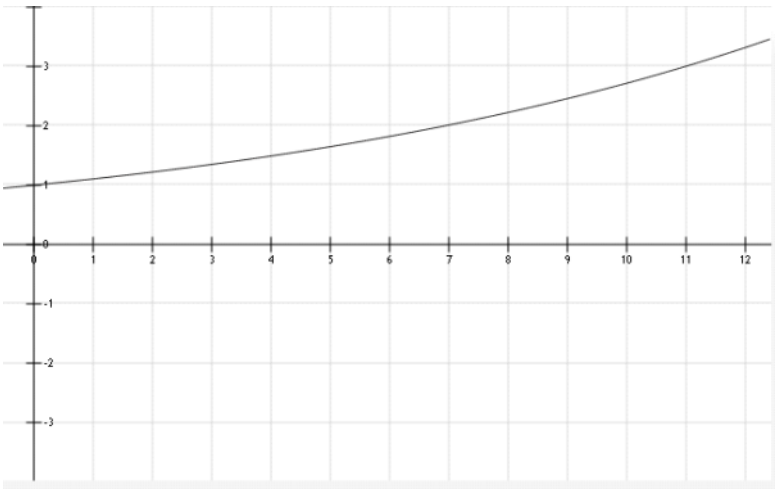
$$A = \frac{y_0, y_1, y_2, \dots\dots, y_n}{n}$$

What about the average value of a function on an interval [a,b] ?





We could approximate by taking a partition of our interval….



Sec 6.5 (Average Value of A Function)

$$f_{avg} \approx \frac{1}{b-a}\sum_{i=1}^{n} f(x_i)\Delta x_i$$

Ex. Find the average value of $f(x) = e^{\frac{1}{10}x}$ on (1,10)

Sec 6.5 (Average Value of A Function)

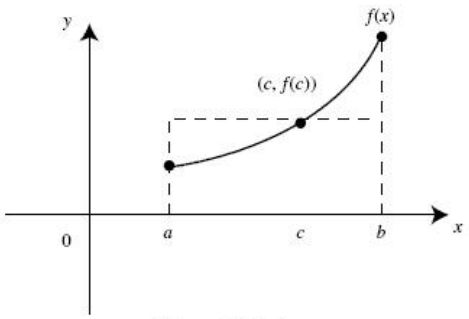
HW Example. Find the average value of $h(x) = \cos^4(x)\sin(x)\ \ on\ [0, \pi]$

Sec 6.5 (Average Value of A Function)

If f is continuous on [a,b] then there exists a number c such that:

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$$

That is; $\int_a^b f(x)dx = f(c)(b-a)$



To prove this recall the mean value theorem for derivatives $f'(c) = \frac{f(b)-f(a)}{b-a}$

Let's apply it to the function $F(x) = \int_a^x f(t)dt$

Sec 6.5 (Average Value of A Function)

Example: Find C such that $f_{avg} = f(c)$ for $f(x) = \ln(x)\ on\ [1,3]$

Sec 6.5 (Average Value of A Function)

Ex. In a certain city the temperature in degrees F is t hours after 9 am is modeled by the function

$$T(t) = 50 + 14\sin\left(\frac{\pi t}{12}\right)$$

Find the average temperature from 9 am to 9 pm

Sec 6.5 (Average Value of A Function)

The velocity of blood that flows in a blood vessel with radius R and length l at a distance r from the central axis is

$$v(r) = \frac{P}{4\eta l}(R^2 - r^2)$$

Where P is the pressure difference between the ends of the vessel and η is the viscosity of the blood. Find the average velocity (with respect to r) on the interval $0 \leq r \leq R$